AIdrona - Detailed Explanation with System Roles

Introduction

AIdrona is the AI-powered medical intelligence system inside our application. Its main purpose is to ensure that only medically eligible users are allowed to donate blood during emergencies.

Why AIdrona is Needed

Without AIdrona, the system would send emergency requests to all users, including those who are not medically fit. This can be dangerous and inefficient. With AIdrona, only safe and eligible donors receive alerts.

What AIdrona Stores

- Medical conditions (e.g., diabetes, blood pressure)
- Medications currently being taken
- Doctor prescriptions (image or PDF)
- Hemoglobin level
- Last blood donation date

How AIdrona Works

Step 1: Data Storage
Users upload their medical data into the system.

Step 2: AI Analysis
The AI checks hemoglobin levels, donation history, diseases, and medications.

Step 3: Eligibility Decision
The system classifies users as eligible, temporarily not eligible, or not eligible.

Step 4: Fitness Score
A score is generated to rank donors based on health and reliability.

How It Works in Emergencies

1. A user raises a blood request.
2. The system finds nearby users.

3. AIdrona analyzes each user's medical data.
4. Only eligible users receive the emergency popup.

Real-Life Example

If a user has low hemoglobin or recently donated blood, AIdrona will block them from receiving alerts. A healthy user will receive the alert.

Simple Analogy

AIdrona acts like a doctor inside the app who checks every user before allowing them to donate.

Role of Hospitals

- Hospitals register through controlled access.
- They update vaccine stock availability.
- They verify donors during donation.
- They mark donors as fit or unfit.
- They update vaccine dose records.

Role of Blood Banks

- Blood banks register through admin access.
- They update blood availability by blood group.
- They act as backup when no donors respond.
- They provide real-time blood stock information.

Importance

- Ensures safety
- Improves efficiency
- Speeds up response

Conclusion

AIdrona makes the system intelligent, safe, and effective by using AI to filter and select the right donors, while hospitals and blood banks ensure resource availability and verification.